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# Attention Is All You Need

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## Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely.

## 1 Introduction

Recurrent neural networks, especially long short-term memory and gated recurrent neural networks, have been firmly established as state of the art approaches in sequence modeling. Recurrent models typically factor computation along the symbol positions of the input and output sequences. This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths. In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output.

## 3 Model Architecture

The Transformer follows this overall architecture using stacked self-attention and point-wise, fully connected layers for both the encoder and decoder. The encoder is composed of a stack of N = 6 identical layers. Each layer has two sub-layers. The first is a multi-head self-attention mechanism, and the second is a simple, position-wise fully connected feed-forward network. We employ a residual connection around each of the two sub-layers, followed by layer normalization. Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions.

## 5 Results

On the WMT 2014 English-to-German translation task, the big transformer model outperforms the best previously reported models by more than 2.0 BLEU, establishing a new state-of-the-art BLEU score of 28.4. On the WMT 2014 English-to-French translation task, our big model achieves a BLEU score of 41.8, outperforming all previously reported single models, at 1/4 the training cost of the previous state-of-the-art model.

## 6 Limitations & Complexity

While self-attention connects all positions with a constant number of sequentially executed operations, its computational complexity per layer is O(n^2 * d), where n is the sequence length and d is the representation dimension. When the sequence length n is very large, self-attention layers can become computationally and memory prohibitive compared to restricted recurrent layers.